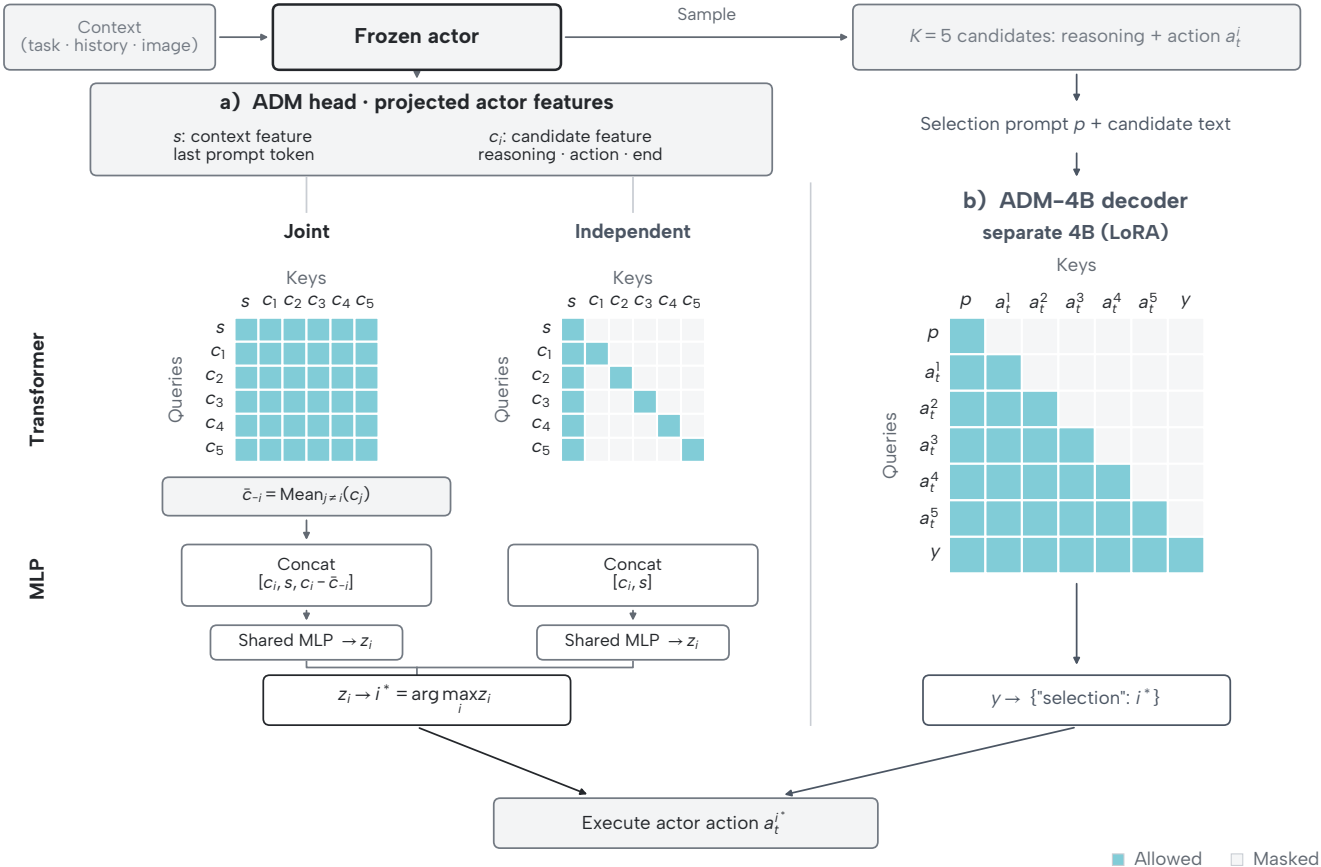
Context
(task · history · image)
Frozen actor
Sample
K = 5 candidates: reasoning + action a_t^i
a) ADM head · projected actor features
s: context feature
last prompt token
c_i: candidate feature
reasoning · action · end
Selection prompt p + candidate text
b) ADM-4B decoder
separate 4B (LoRA)
Joint
Independent
Transformer
MLP
Keys
Queries
s c1 c2 c3 c4 c5
p a_t^1 a_t^2 a_t^3 a_t^4 a_t^5 y
$\bar{c}_{-i} = \text{Mean}_{j \neq i}(c_j)$
Concat
$[c_i, s, c_i - \bar{c}_{-i}]$
Concat
$[c_i, s]$
Shared MLP → z_i
Shared MLP → z_i
$z_i \rightarrow i^* = \arg\max_i z_i$
$y \rightarrow$ {"selection": i^*}
Execute actor action $a_t^{i^*}$
Allowed
Masked